A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 - 2026

Class: T.E. A Sem: VI

Project Title: Krushi-Yantra: Gamified, Climate-Safe Farming with AI & Carbon Income

Team Member 1: (Team Leader)

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Team Member 3:

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Name: Rajanya Sandeep Kshatriya

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Prof. Dr. Tejashri Kolhe — Project Guide

Prof. Yogesh Hardas — Mini- Project Coordinator

Dr. _____ — Head of Department

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 09/01/2026 to 15/01/2026

Progress Planned	Progress Achieved
1) Define System Architecture	1) Architecture finalised
2) Set up GitHub repo with branch strategy	2) GitHub repo live
3) Draft API contract for Farmer & Admin routes	3) API endpoint schemas agreed for all routes

Guides Review: Architecture finalized, start work on cypher scripts

Signature

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

Name and Signature of Team Member 3:

Name and Signature of Team Member 4:

Name of Project guide: Dr. T.T.K

Signature of guide with Date: 15/1/26

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 16/01/2026 to 22/01/2026

Progress Planned	Progress Achieved
1) Design Neo4j Schema	1) Graph schema done including zone polygones
2) Write Cypher Scripts for forms	2) 5 forms seeded across district
3) Test neighbourhood traversal query by soil type.	3) Neighbour traversal live

Guides Review: Start work on FastAPI, registration.

Signature

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

Name and Signature of Team Member 3:

Name and Signature of Team Member 4:

Name of Project guide: Dr. JTK

Signature of guide with Date: 22/1/26

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 23/01/2026 to 29/01/2026

Progress Planned	Progress Achieved
1) Build FastAPI scaffold with farmer, admin, auth routers	1) FastAPI scaffold complete.
2) Create farmer registration endpoints	2) Farmer registration live in Neo4j

Guides Review: Fast API completed., work assigned completed.

Signature

Name and Signature of Team Member 1:

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Name and Signature of Team Member 4:

Name of Project guide: Dr. J. Jk

Signature of guide with Date: 29/1/26

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 30/01/2026 to 05/02/2026

Progress Planned	Progress Achieved
1) Build form polygon API accepting GeoJSON	1) GeoJSON zone endpoint live
2) Use shapely for area calculation & geo-conflict check.	2) Geo-conflict detection working.

Guides Review: polygon API endpoint created.

Signature

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

Name and Signature of Team Member 3:

Name and Signature of Team Member 4:

Name of Project guide: Dr. TTK

Signature of guide with Date: 5/2/26

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 06/02/2026 to 12/02/2026

Progress Planned	Progress Achieved
1) Build crop recomendation API via Neo4j graph traversal	1) Graph traversal returning neighbour-validated crops
2) Integrate Open-Meteo 16-day weather forecast	2) Open-Meteo integrated
3) Return top 3 crops with profit, cost & water requirement.	3) Top 3 crops returning with profit.

Guides Review: Integrated open-meteo, worked on crops

Signature

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

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Name and Signature of Team Member 4:

Name of Project guide: Dr. JJK

Signature of guide with Date: 12/2/26

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 13/02/2026 to 19/02/2026

Progress Planned	Progress Achieved
1) Load YOLOv8 (PlantVillage)	1) YOLOv8 serving
2) Pass YOLO crop to Gemini vision for disease analysis	2) Gemini returns disease name, spread risk.
3) Implement 10km disease alert system.	3) 10km alert query built on Neo4j

Guides Review: yolo V8 used, worked on distance.

Signature

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Name and Signature of Team Member 4:

Name of Project guide: Dr. TTK

Signature of guide with Date: 19/2/26

Parshvanath Charitable Trust's
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 20/02/2026 to 26/02/2026

Progress Planned	Progress Achieved
1) Configure Twilio Webhook with Hindi speech-to-text	1) Twilio webhook configured
2) Generate Hindi TTS responses	2) TTS 20 queries per-cached for offline mode.

Guides Review: Twilio configured.

Signature

Name and Signature of Team Member 1:

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Name and Signature of Team Member 4:

Name of Project guide: Dr. T. J. K

Signature of guide with Date: 26/2/26

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 27/02/2026 to 05/03/2026

Progress Planned	Progress Achieved
1) Build admin role-based JWT	1) Admin JWT working
2) Create admin overview with aggregate stats from Neo4j	2) Overview endpoint returning live.

Guides Review: Created. JWT admin and interface with Neo4j

Signature :

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

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Name and Signature of Team Member 4:

Name of Project guide: Dr. T.T.

Signature of guide with Date: 5/3/26

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 06/03/2026 to 12/03/2026

Progress Planned	Progress Achieved
1) Set up LangChain agent with Neo4j	1) LangChain Neo4j toolkit integrated
2) Test 10 admin queries	2) Conversation history buffer added.

Guides Review: Integration of Long chain Neo4j with toolkit done.

Signature

Name and Signature of Team Member 1:

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Name and Signature of Team Member 4:

Name of Project guide: Dr. TTK

Signature of guide with Date: 12/3/26

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 13/03/2026 to 19/03/2026

Progress Planned	Progress Achieved
1) Build disease outbreak map endpoint	1) Disease outbreak endpoint returning alert nodes.
2) Build carbon credit logic	2) Carbon credit calculator live

Guides Review: calculation of carbon credit done.

Signature

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Name of Project guide: Dr. T.J.K

Signature of guide with Date: 19/3/26

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 20/03/2026 to 26/03/2026

Progress Planned	Progress Achieved
1) Gate credit activation on form document approve status 2) Build admin collective pool endpoint	1) Credit activation gated correctly 2) Admin pool endpoint aggregating

Guides Review: Activated. gate credit.. and. admin aggregation done

Signature

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

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Name and Signature of Team Member 4:

Name of Project guide: Dr. TTK

Signature of guide with Date: 26/3/25

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 27/03/2026 to 02/04/2026

Progress Planned	Progress Achieved
1) Run end-to-end tests	1) E2E tests done. 3 bugs fixed
2) Deploy FastAPI to Railway	2) Backend deployed on Railway

Guides Review: Backend was deployed on Railway.

Signature

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

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Name and Signature of Team Member 4:

Name of Project guide: Dr. JJK

Signature of guide with Date: 2/4/26

Gaikwad Mumbai Trust's

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Academic Year: 2025 – 2026

Class: T.E. A Sem: VI

Date: From 03/04/2026 to 09/04/2026

Progress Planned	Progress Achieved
1) Prepare demo data	1) Demo data loaded
2) Changing some UI components	2) All UI components were changed as per requirement
3) Whole system run smothly	3) System runing smoothly.

Guides Review: System working. Project work Completed.

Signature

Name and Signature of Team Member 1:

Name and Signature of Team Member 2:

Name and Signature of Team Member 3:

Name and Signature of Team Member 4:

Name of Project guide: Dr. TTK

Signature of guide with Date: 9/4/26